



nlmixr2save

Fits you can read, runs you don't repeat

Change one operator: `fit := nlmixr2(model, data)`

Saving and caching nlmixr2 fits: portable text files and the := operator

Objectives: Nonlinear mixed-effects (NLME) estimation and simulation take minutes to weeks. In reproducible workflows, re-running models unnecessarily wastes time, and language-specific binary serialization limits sharing results. nlmixr2[1] relied on serialization that was removed from the R ecosystem, making this urgent. We aimed to (1) save nlmixr2 fits as primarily CSV and R source files, and (2) add a caching operator, :=, that reruns only when the model, data, or settings changed.

Methods: nlmixr2save was developed within the nlmixr2 ecosystem in R. Fits are decomposed into portable components – the final estimated model (as nlmixr2/rxode2[1–2] code), objective function values, tables, and metadata – written as R and CSV text and reconstructed without binary dependencies. The := operator is a drop-in replacement for <- when calling nlmixr2(), rxSolve(), or any long-running R function. It hashes the model, the estimation-relevant data columns, and all estimation or simulation options, plus the starting seed for random calls. A matching cache is loaded from disk and, for random calls, the final seed state restored; otherwise the call runs and the result is saved.

Results: Fits round-trip through a single zip file. Loaded fits keep full functionality: diagnostic plots, VPCs, and use as initial estimates for new runs. Switching `fit <- nlmixr2(model, data, est="saem")` to `fit := nlmixr2(model, data, est="saem")` is the only change needed for caching. Other functions are cached in rds format. Version 0.2.0 adds cache directory and prefix options, package-version checks, and nlmixr2saveShare() to share a fit without the original data.

Conclusions: nlmixr2save addresses two practical problems: long-term readable storage of nlmixr2 fits, and redundant runs during iterative development. The := operator gives intelligent result management with minimal effort, promoting efficient, reproducible, collaborative modeling – especially when scripts are re-sourced or rendered in dynamic documents.

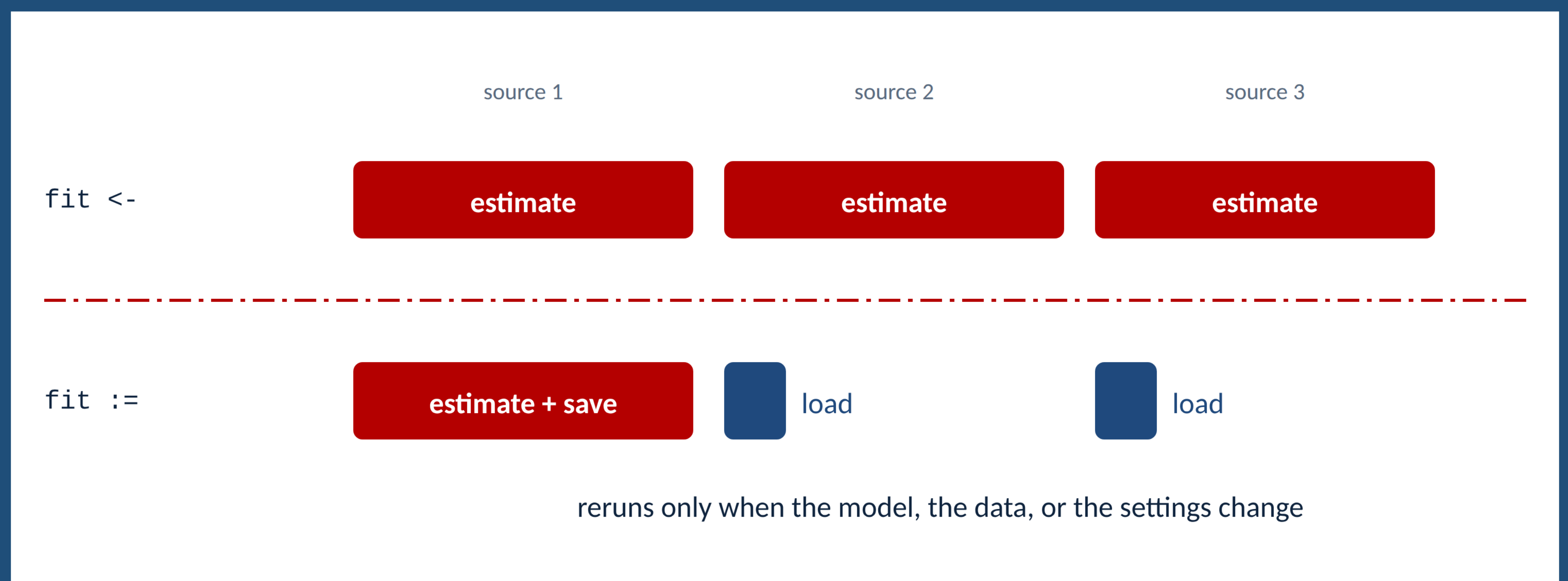
References

[1] Fidler M, et al. CPT Pharmacometrics Syst Pharmacol. 2019;8(9):621–633.

[2] Wang W, Hallow KM, James DA. CPT Pharmacometrics Syst Pharmacol. 2016;5(1):3–10.

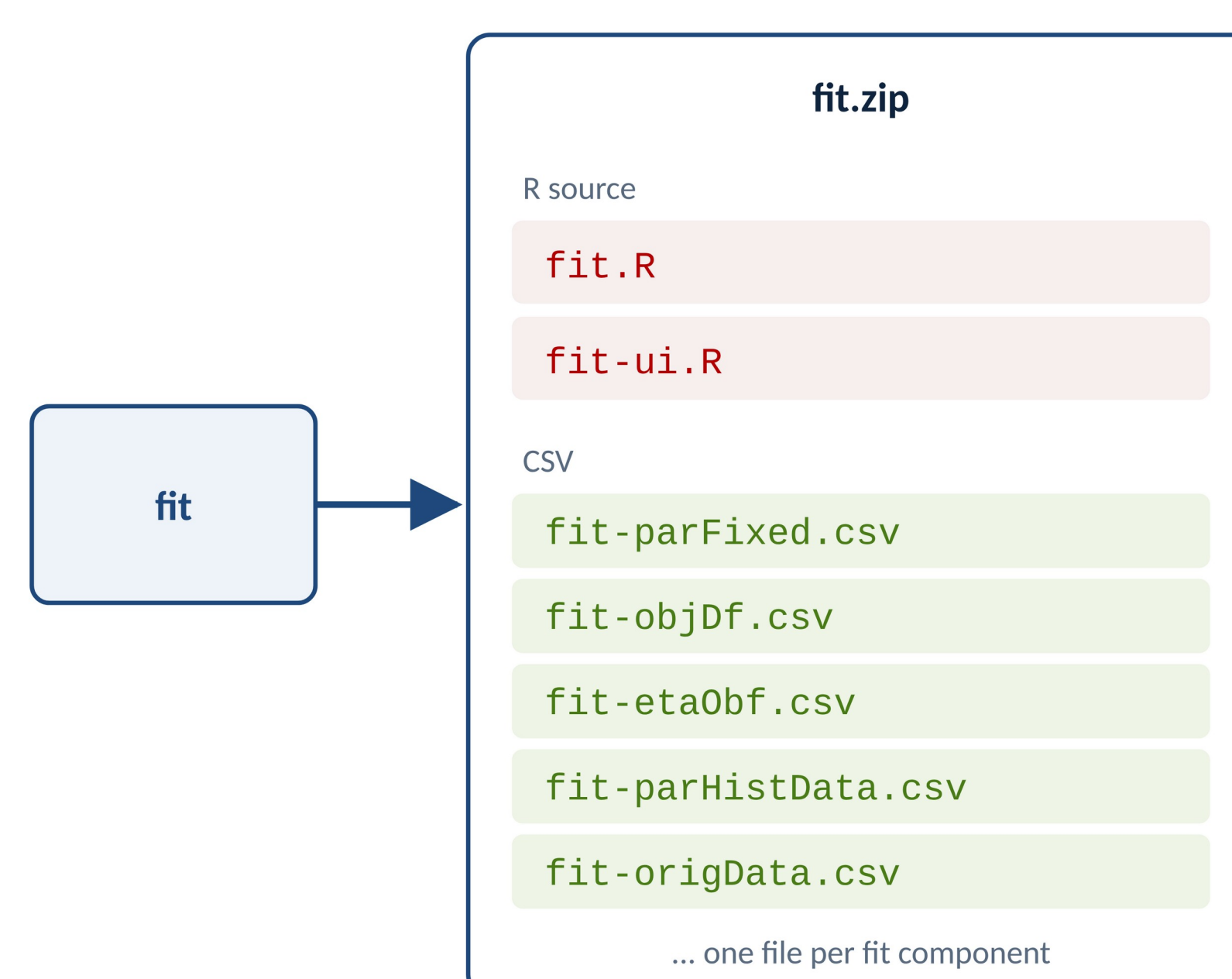
Install:

`install.packages("nlmixr2save")`

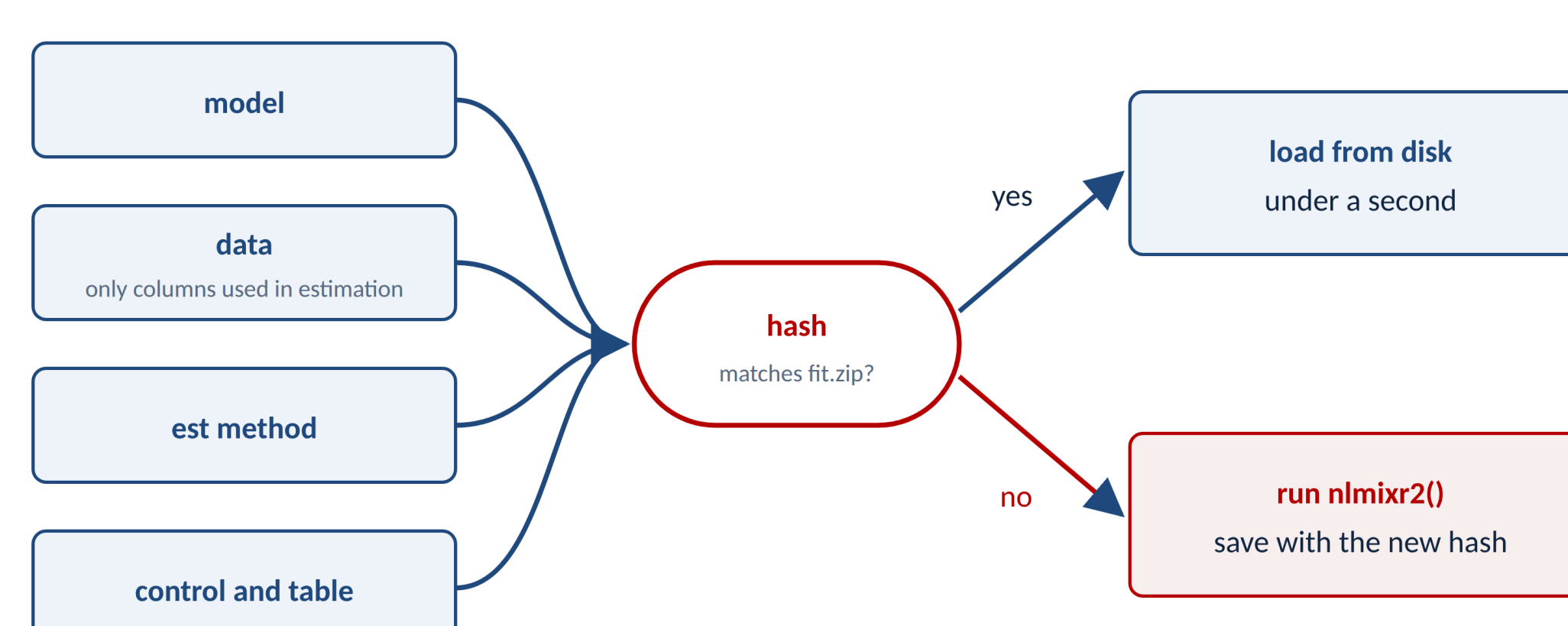


Re-sourcing a script or re-rendering a report reruns nothing **unless the model, data, or settings changed**

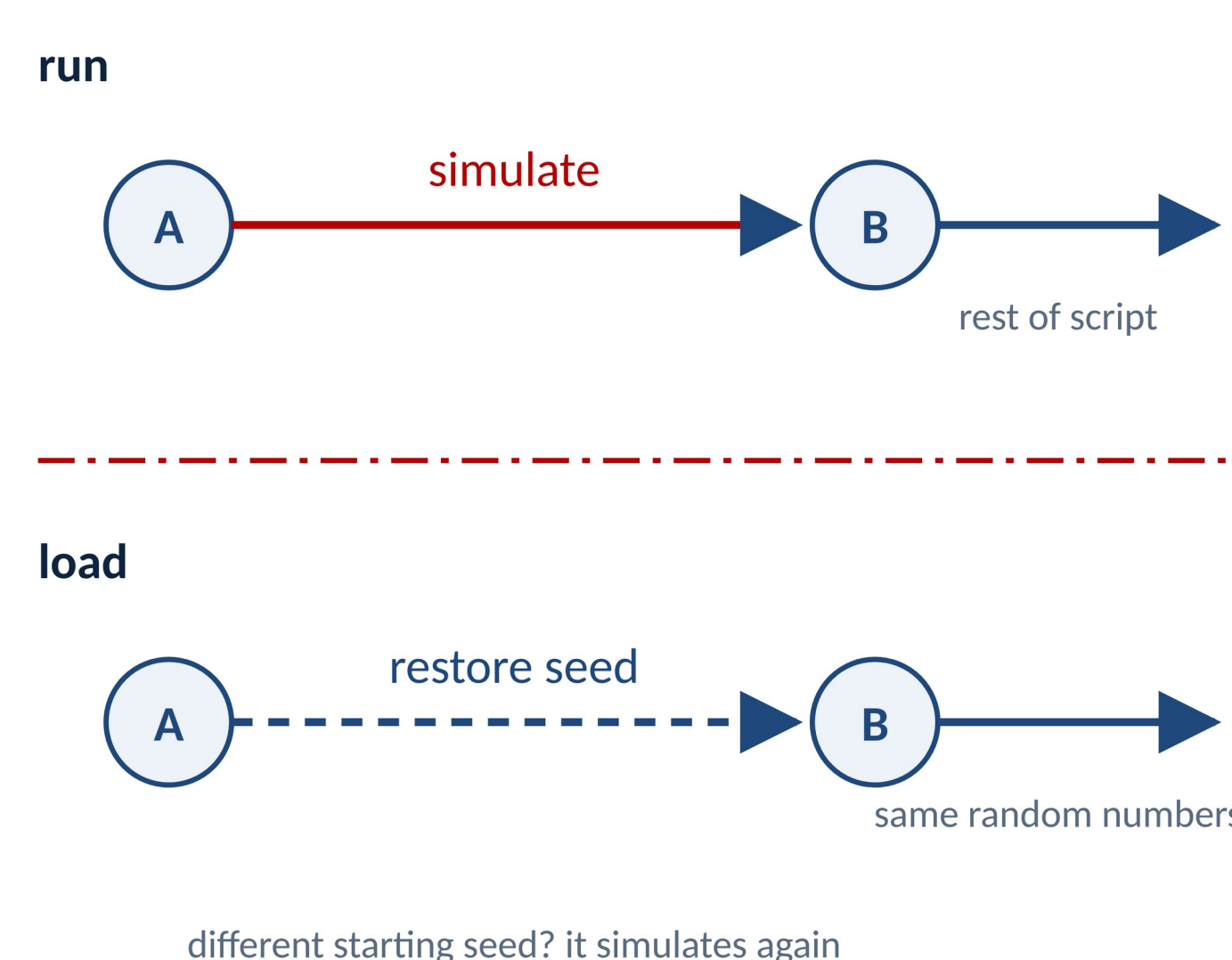
saveFit() writes R source and CSV into one zip



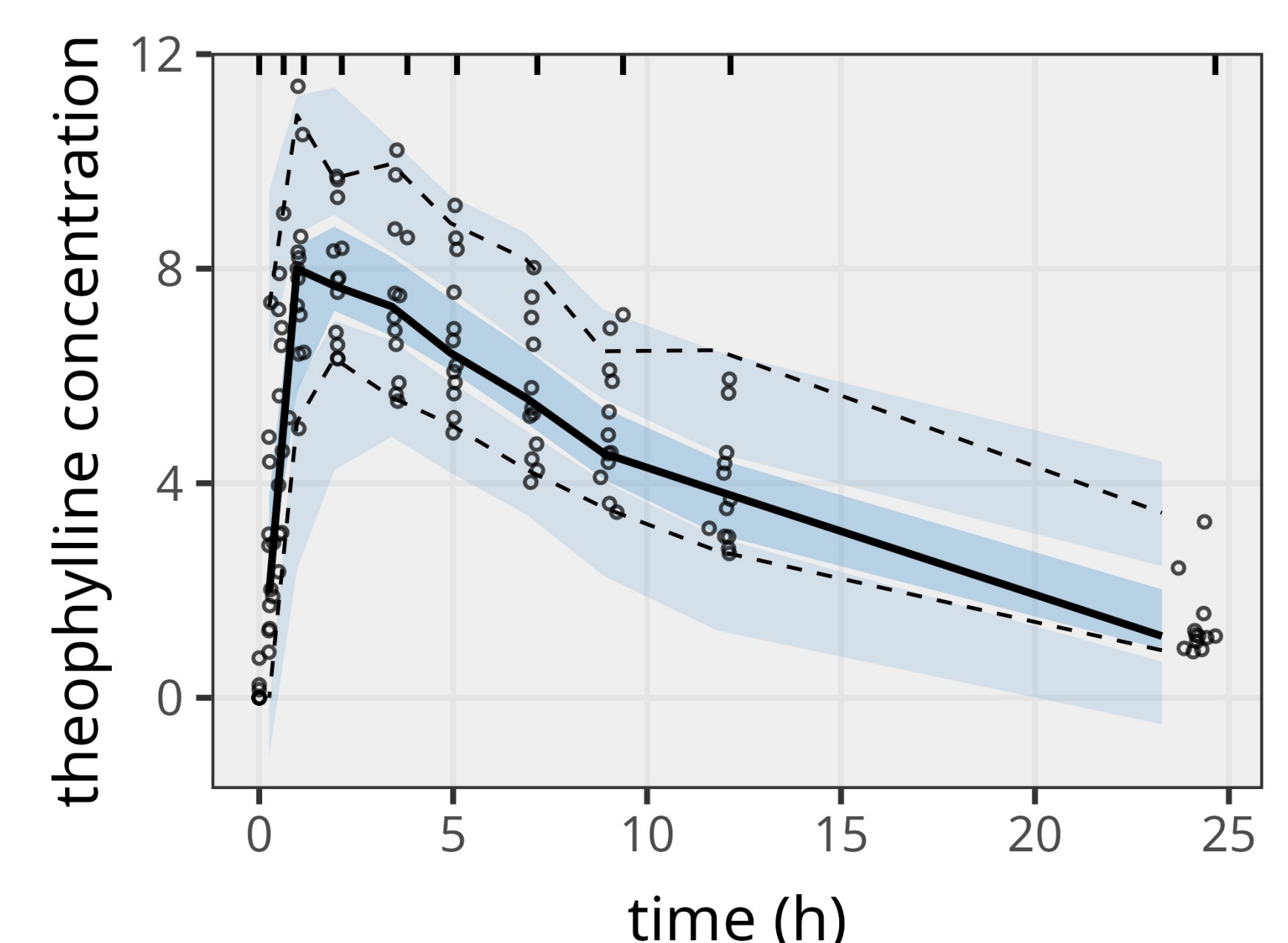
The cache key hashes what the estimate depends on



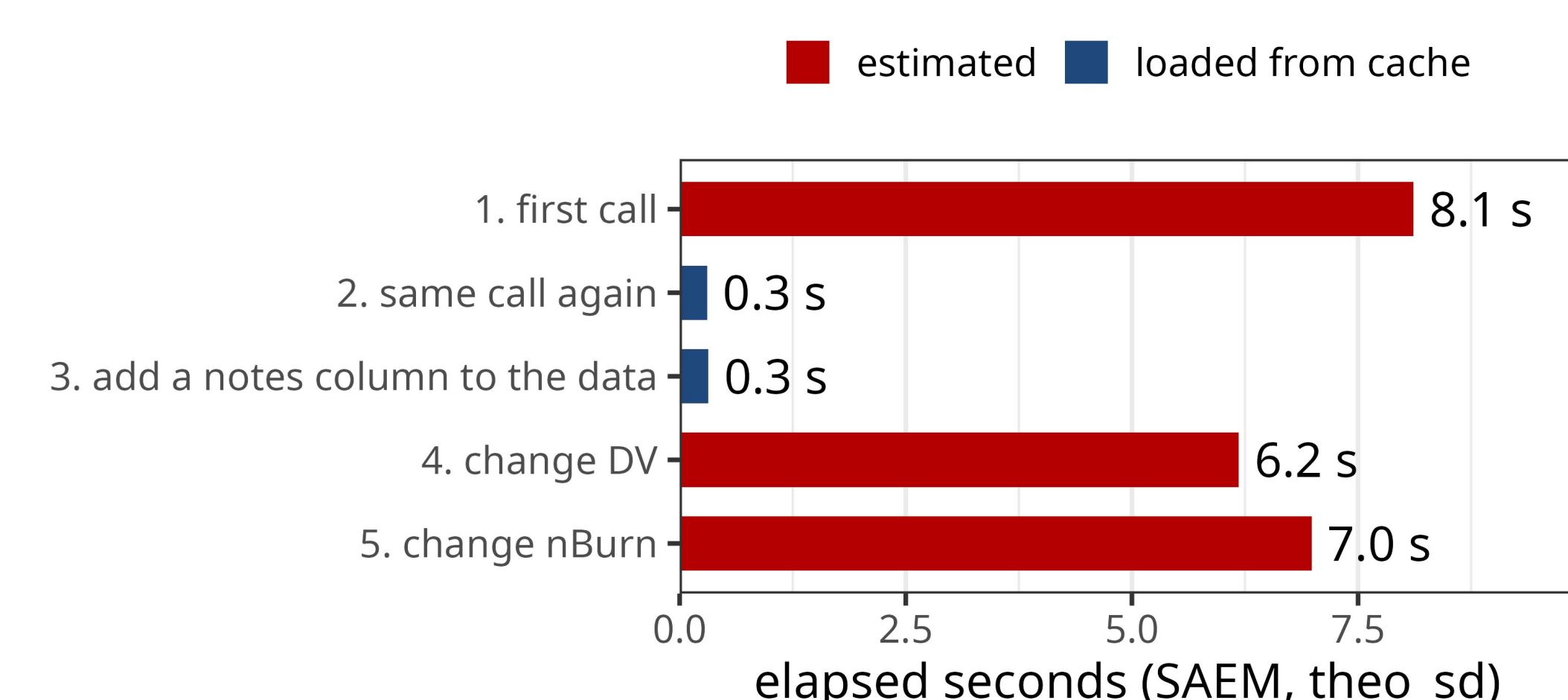
Simulations restore the random seed



A loaded fit still makes VPCs



Only changes that matter trigger a refit



nlmixr2saveShare(): share a fit without the data

	-noData	-noData-noFit
model, estimates, objective	kept	kept
etas, parameter history	kept	kept
predictions, residuals	kept	removed
original data	removed	removed

fit.zip is left unchanged; copies are written next to it

Authors

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Blog post + video



Documentation